

A product counterexample to the simplex maximum for projection-body volume

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Abstract

The product of two ten-dimensional simplices has larger normalized projection-body volume than a twenty-dimensional simplex. This gives a counterexample to Brannen’s proposed simplex maximum.

1 Introduction

Throughout, a convex body in $\mathbb{R}^d$ is compact and convex with nonempty interior, and $d \geq 2$. Write $|K|$ for its d -dimensional volume and $h_C(x) = \max_{y \in C} \langle x, y \rangle$ for the support function of a compact convex set C . The projection body $\Pi_d K$, also written ΠK when the dimension is evident, is defined by

$$h_{\Pi K}(u) = \text{vol}_{d-1}(\text{proj}_{u^\perp} K), \quad u \in S^{d-1},$$

extended homogeneously to all of $\mathbb{R}^d$. Here S^{d-1} is the Euclidean unit sphere and $\text{proj}_{u^\perp}$ is orthogonal projection. We write B_2^j for the Euclidean unit ball in $\mathbb{R}^j$ and $\kappa_j = |B_2^j|$. The functional under consideration is

$$R_d(K) = \frac{|\Pi K|}{|K|^{d-1}}, \quad c_d = \frac{(d+1)d^d}{d!}. \quad (1)$$

The extremal values of this affine-invariant functional are classical questions in convex geometry. Petty established the affine covariance of the projection-body operator [6]; see also [5, Equation (1)]. Brannen conjectured in 1996 that simplices maximize R_d among all convex bodies [1]. The simplex value is c_d , as we verify below. Thus the proposed simplex upper bound is

$$|\Pi K| \leq c_d |K|^{d-1}. \quad (2)$$

It is the volume of ΠK itself that occurs here, not the volume of its polar: the classical projection inequalities involving the polar concern a different functional. For $d \geq 3$, Petty’s separate minimum problem asks whether ellipsoids are the only minimizers of the same functional R_d ; see [7, Section 1] for these distinctions.

In dimension three, Chen, Feng, Li, Xi, and Xu settle the minimum problem and also prove the bound $R_3(K) \leq 18 = c_3$ for all three-dimensional convex bodies, with tetrahedra attaining equality [2, Theorems 1.1 and 1.2]. Feng, Hu, Liu, and Xu give counterexamples to (2) in every dimension $d \geq 9$, using polytopes with at most $d+2$ facets [3, Theorem 1.3]. Their result already gives a negative answer to the unrestricted-dimensional simplex-maximum question. Their construction uses perturbed projections of a cube and Minkowski’s existence theorem [3, Section 4]. The present paper supplies a direct Cartesian-product construction and its exact value: two ten-dimensional simplices suffice.

Theorem 1. *Let $T_{10} = \text{conv}(0, e_1, \dots, e_{10}) \subset \mathbb{R}^{10}$, where the e_i are the coordinate unit vectors, and let $K = T_{10} \times T_{10} \subset \mathbb{R}^{20}$. Then*

$$\frac{R_{20}(K)}{c_{20}} = \frac{121 \binom{20}{10}}{21 \cdot 2^{20}} = \frac{22\,355\,476}{22\,020\,096} > 1.$$

Thus K violates (2) in dimension twenty.

The exact witness in Theorem 1 provides an elementary illustration of the failure of the simplex maximum. The optimal upper constant and its maximizing bodies remain separate questions.

The useful mechanism is a product test: for full-dimensional polytopes $A \subset \mathbb{R}^r$ and $B \subset \mathbb{R}^s$, with $r, s \geq 2$,

$$R_{r+s}(A \times B) = R_r(A)R_s(B).$$

Consequently, the product $c_r c_s$ of the simplex values is a necessary lower bound for any universal upper constant in dimension $r + s$. The dimension-twenty counterexample comes from comparing two copies of the ten-dimensional simplex value with the twenty-dimensional simplex value. All identities used in that comparison are proved below.

The growth of upper constants with dimension is a further quantitative question. For centrally symmetric bodies, Henk gives the lower bound $2^d(9/8)^{\lfloor d/3 \rfloor}$ for the optimal upper constant [4]. Iterating our product identity gives an exponential excess over the simplex benchmark: for some fixed $\lambda > 1$ and every sufficiently large n , a product of simplices $K_n \subset \mathbb{R}^n$ satisfies $R_n(K_n) \geq \lambda^n c_n$. Corollary 6 proves this consequence by repeating ten-dimensional factors and using one additional simplex to reach each dimension.

Section 2 derives the facet formula, the product identity, and affine covariance. Section 3 computes the simplex value by expressing its projection body as a cube plus one segment. Section 4 combines the identities and gives the exact integer comparison, then proves the exponential excess in all sufficiently large dimensions.

2 Facet vectors and Cartesian products

We first express projection bodies of polytopes as sums of segments. The facets of a Cartesian product then separate into two families, which will make its normalized projection-body volume multiplicative. The facet formula is the polytopal case of Cauchy's projection formula; see [7, Section 1]. We include its proof to fix the factor $1/2$ on which the subsequent constants depend.

We use Minkowski addition $C + D = \{c + d : c \in C, d \in D\}$. Support functions add under this operation and determine compact convex sets uniquely. For the latter assertion, if $x \notin C$ and $y \in C$ is nearest to x , minimization along each segment $[y, z] \subset C$ gives $\langle x - y, z - y \rangle \leq 0$ for every $z \in C$. The linear functional with vector $x - y$ therefore separates x from C .

Lemma 2. *Let $P \subset \mathbb{R}^d$, $d \geq 2$, be a full-dimensional convex polytope. For each facet F , let s_F be its $(d - 1)$ -dimensional volume and ν_F its outward unit normal. Then*

$$h_{\Pi P}(u) = \frac{1}{2} \sum_F s_F |\langle \nu_F, u \rangle|, \quad u \in \mathbb{R}^d, \quad (3)$$

and therefore

$$\Pi P = \sum_F \left[-\frac{s_F \nu_F}{2}, \frac{s_F \nu_F}{2} \right]. \quad (4)$$

Proof. First take u to be a unit vector. Orthogonal projection from the hyperplane of F to $u^\perp$ has absolute Jacobian $|\langle \nu_F, u \rangle|$. Indeed, for an orthonormal tangent basis $v_1, \dots, v_{d-1}$, that Jacobian is $|\det(v_1, \dots, v_{d-1}, u)|$, which equals the stated inner product in absolute value.

Apart from a set of $(d-1)$ -dimensional measure zero, a point in the projection of P lies on the projections of exactly two facet interiors: the entry and exit points of its line parallel to u . To see that the exceptions are null, discard the boundary of the projection and the projections of all faces of dimension at most $d-2$. Also discard projections of facets whose hyperplanes are parallel to u ; these projections have dimension at most $d-2$. Every remaining line meets P in a nondegenerate interval with its two endpoints in facet interiors. Integrating this multiplicity gives (3) for unit u , and homogeneity gives the general case.

The support function of $[-v/2, v/2]$ is $\frac{1}{2}|\langle v, u \rangle|$. Adding these support functions proves (4). $\square$

Proposition 3. *Let $A \subset \mathbb{R}^r$ and $B \subset \mathbb{R}^s$ be full-dimensional convex polytopes, where $r, s \geq 2$. In the orthogonal product space $\mathbb{R}^{r+s} = \mathbb{R}^r \times \mathbb{R}^s$,*

$$\Pi(A \times B) = (|B|\Pi A) \times (|A|\Pi B). \quad (5)$$

In particular,

$$R_{r+s}(A \times B) = R_r(A)R_s(B). \quad (6)$$

Proof. The face exposed by a linear functional (u, v) on $A \times B$ is the product of the faces exposed by u and v . Dimensions add under Cartesian products. A nonzero linear functional exposes a proper face of a full-dimensional body, so if both u and v are nonzero, the product face has codimension at least two. A facet therefore has one facet factor and one whole factor. Thus the facets are $F \times B$ and $A \times G$, with area-normal vectors

$$(|B|s_F\nu_F, 0) \quad \text{and} \quad (0, |A|s_G\nu_G),$$

respectively. The area factors are products because the two coordinate spaces are orthogonal. Lemma 2 now gives

$$h_{\Pi(A \times B)}(u, v) = |B|h_{\Pi A}(u) + |A|h_{\Pi B}(v).$$

This is the support function of the right side of (5), proving that identity. Hence

$$\begin{aligned} R_{r+s}(A \times B) &= \frac{|B|^r |A|^s |\Pi A| |\Pi B|}{(|A||B|)^{r+s-1}} \\ &= \frac{|\Pi A|}{|A|^{r-1}} \frac{|\Pi B|}{|B|^{s-1}}, \end{aligned}$$

as required. $\square$

The product identity reduces our comparison to the values of its factors. To use simplices as a dimension-dependent benchmark, we also need to know that all full-dimensional simplices of a fixed dimension have the same value. The following affine covariance formula supplies that fact.

Lemma 4. *Let $P \subset \mathbb{R}^d$, $d \geq 2$, be a convex body. For every invertible linear map $L: \mathbb{R}^d \rightarrow \mathbb{R}^d$,*

$$\Pi(LP) = |\det L|L^{-t}\Pi P, \quad R_d(LP) = R_d(P),$$

where L^{-t} denotes the inverse transpose. Translations also preserve R_d .

Proof. First suppose that P is a polytope. The transformed outward unit normal of F is $L^{-t}\nu_F/\|L^{-t}\nu_F\|$. Its area is

$$s_{LF} = |\det L| \|L^{-t}\nu_F\| s_F.$$

For the area formula, take a prism of height one over F , extending in direction ν_F . Its image has volume $|\det L|s_F$ and height $1/\|L^{-t}\nu_F\|$ over LF ; dividing volume by height gives the formula. Thus each area-normal vector transforms by $|\det L|L^{-t}$, including when $\det L < 0$. Apply (4). The absolute determinant of this map is $|\det L|^{d-1}$, exactly the scaling of $|LP|^{d-1}$. Translation invariance follows directly from projection volumes.

For completeness, we justify the passage to an arbitrary convex body and the existence of its projection body. Translate an interior point of P to the origin, choose $R > 0$ with $P \subset RB_2^d$, and take increasing inscribed polytopes P_m containing a fixed ball aB_2^d , $a > 0$, such that

$$P_m \subset P \subset P_m + \varepsilon_m B_2^d \subset (1 + \varepsilon_m/a)P_m, \quad \varepsilon_m \rightarrow 0.$$

Take, for example, convex hulls of cumulative finite nets in P and a fixed inscribed polytope containing aB_2^d . Put $\lambda_m = 1 + \varepsilon_m/a$, and for unit u write $F(u) = \text{vol}_{d-1}(\text{proj}_{u^\perp} P)$ and $f_m(u) = h_{\Pi P_m}(u)$. Projection and volume monotonicity give

$$0 \leq F(u) - f_m(u) \leq (\lambda_m^{d-1} - 1)\kappa_{d-1}R^{d-1}.$$

Thus $f_m \rightarrow F$ uniformly. The increasing bodies ΠP_m lie in a fixed ball and contain $\kappa_{d-1}a^{d-1}B_2^d$, so their union closure is a convex body with support function F , namely ΠP . The support inequalities also give

$$\Pi P_m \subset \Pi P \subset \lambda_m^{d-1}\Pi P_m, \quad |\Pi P_m| \leq |\Pi P| \leq \lambda_m^{d(d-1)}|\Pi P_m|.$$

Likewise $|P_m| \leq |P| \leq \lambda_m^d|P_m|$. Applying L to the original sandwich gives the same convergence for LP_m . Passing to the limit in the polytope identity proves covariance and, by these volume inequalities, the asserted invariance of R_d . $\square$

3 The simplex value

We now compute the value to which the product will be compared. Only one extra segment beyond a coordinate cube is needed to describe the projection body of a simplex.

Proposition 5. *For $d \geq 2$, the standard simplex $T_d = \text{conv}(0, e_1, \dots, e_d) \subset \mathbb{R}^d$ satisfies*

$$|T_d| = \frac{1}{d!}, \quad |\Pi T_d| = \frac{d+1}{((d-1)!)^d}, \quad R_d(T_d) = c_d.$$

Every full-dimensional d -simplex has the same value c_d .

Proof. The simplex volume follows by iterated base and height. Put $w = e_1 + \dots + e_d$. The coordinate facets have area-normal vectors $-e_i/(d-1)!$. The remaining facet has unit normal $w/\sqrt{d}$. Dropping its last coordinate projects it onto the standard $(d-1)$ -simplex with Jacobian $1/\sqrt{d}$; its area-normal vector is therefore $w/(d-1)!$. In the centered segments of Lemma 2, changing the sign of a generator changes no segment. Translating each segment to start at the origin therefore shows that ΠT_d is a translate of

$$\frac{[0, 1]^d + [0, w]}{(d-1)!}.$$

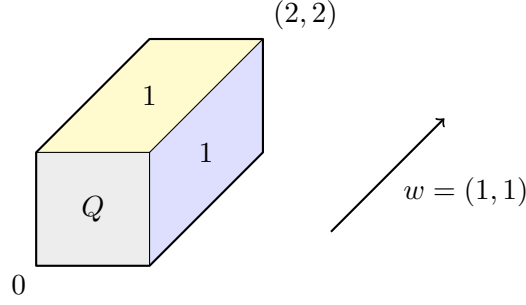


Figure 1: The planar extrusion $Q + [0, (1, 1)]$, with $Q = [0, 1]^2$, consists of the unit square and two parallelograms of area one. Its area is 3. The fiber argument in Proposition 5 proves the corresponding volume $d + 1$ in every dimension.

To compute the numerator's volume, let $Q = [0, 1]^d$. Every nonempty fiber of Q parallel to w is an interval, and adding $[0, w]$ increases its length by $\|w\|$. The base projection is unchanged. Fubini therefore gives

$$|Q + [0, w]| = |Q| + \|w\| \text{vol}_{d-1}(\text{proj}_{w^\perp} Q) = 1 + h_{\Pi Q}(w).$$

The cube has two facets of area one in each coordinate direction, so (3) gives $h_{\Pi Q}(w) = d$. Thus $|Q + [0, w]| = d + 1$; Figure 1 illustrates the two-dimensional case. Dilation by $1/(d-1)!$ and division by $|T_d|^{d-1}$ now yield

$$R_d(T_d) = \frac{(d+1)(d!)^{d-1}}{((d-1)!)^d} = \frac{(d+1)d^d}{d!}.$$

Every full-dimensional simplex is an invertible affine image of T_d , so Lemma 4 proves the final assertion. $\square$

4 The counterexample and exponential excess

The geometric work is complete. The product formula turns the comparison in dimension twenty into a calculation with two simplex constants.

Proof of Theorem 1. The product $K = T_{10} \times T_{10}$ is compact and convex with nonempty interior in $\mathbb{R}^{20}$. Propositions 3 and 5 give

$$R_{20}(K) = R_{10}(T_{10})^2 = \left(\frac{11 \cdot 10^{10}}{10!} \right)^2.$$

Consequently,

$$\frac{R_{20}(K)}{c_{20}} = \frac{121 \cdot 10^{20} \cdot 20!}{(10!)^2 \cdot 21 \cdot 20^{20}} = \frac{121 \binom{20}{10}}{21 \cdot 2^{20}} = \frac{22\,355\,476}{22\,020\,096} > 1.$$

Here $\binom{20}{10} = 184\,756$, and the exact integer certificate is

$$121 \cdot 184\,756 - 21 \cdot 1\,048\,576 = 335\,380 > 0.$$

This is an exact strict violation of (2). $\square$

More generally, for $r, s \geq 2$, the same two propositions give the explicit product test

$$\frac{R_{r+s}(T_r \times T_s)}{c_{r+s}} = \frac{(r+1)(s+1)}{r+s+1} \binom{r+s}{r} \frac{r^r s^s}{(r+s)^{r+s}}.$$

Thus any universal upper constant in dimension $r+s$ must be at least $c_r c_s$.

Iterating the product test gives the following quantitative consequence.

Corollary 6. *There exist a constant $\lambda > 1$ and an integer $N \geq 20$ such that, for every integer $n \geq N$, some full-dimensional convex polytope $K_n \subset \mathbb{R}^n$ satisfies*

$$\frac{R_n(K_n)}{R_n(T_n)} \geq \lambda^n.$$

The polytopes K_n may be chosen as Cartesian products of simplices.

Proof. We use repeated ten-dimensional factors to obtain exponential growth, and one factor of dimension between ten and nineteen to reach every sufficiently large dimension. For $r \in \{0, \dots, 9\}$, put $d_r = 10 + r$. Every integer $n \geq 20$ has a unique representation $n = 10k + d_r$ with $k \geq 1$. In the orthogonal coordinate product space, take

$$K_n = T_{10}^k \times T_{d_r}.$$

Here T_{10}^k denotes the k -fold Cartesian product. All factors and their successive products are full-dimensional convex polytopes, with each factor dimension at least ten. Thus Propositions 3 and 5 apply iteratively and give

$$R_n(K_n) = c_{10}^k c_{d_r}.$$

To compare this value with c_n , we bound the simplex constants by $n+1$ times a fixed exponential, whose rate is smaller than that supplied by the ten-dimensional factors. Put $\beta = 11/4$. The exponential series gives

$$e = \frac{5}{2} + \sum_{j=3}^{\infty} \frac{1}{j!} < \frac{5}{2} + \frac{1}{6} \sum_{j=0}^{\infty} 3^{-j} = \beta,$$

since the successive ratios in the tail are strictly less than $1/3$. The positive term $n^n/n!$ in the series for e^n is strictly smaller than the full sum, so

$$c_n < (n+1)e^n < (n+1)\beta^n \quad (n \geq 2).$$

For the fixed ten-dimensional block, exact arithmetic yields

$$c_{10} = \frac{17\,187\,500}{567}, \quad A := \frac{c_{10}}{\beta^{10}} = \frac{1\,638\,400\,000\,000}{1\,336\,956\,340\,797} > 1.$$

Choose a fixed λ with $1 < \lambda < A^{1/10}$. Combining these estimates with the formula for $R_n(K_n)$ gives

$$\frac{R_n(K_n)}{R_n(T_n)\lambda^n} > c_{d_r}\beta^{-d_r}\lambda^{-d_r} \frac{(A/\lambda^{10})^k}{10k + d_r + 1}.$$

For each fixed r , the right side tends to infinity as $k \rightarrow \infty$, because $A/\lambda^{10} > 1$. A threshold can therefore be chosen in each of the ten residue classes; their maximum, enlarged to be at least twenty, gives one N for all $n \geq N$. $\square$

References

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